

CLICK CLICK BOOM

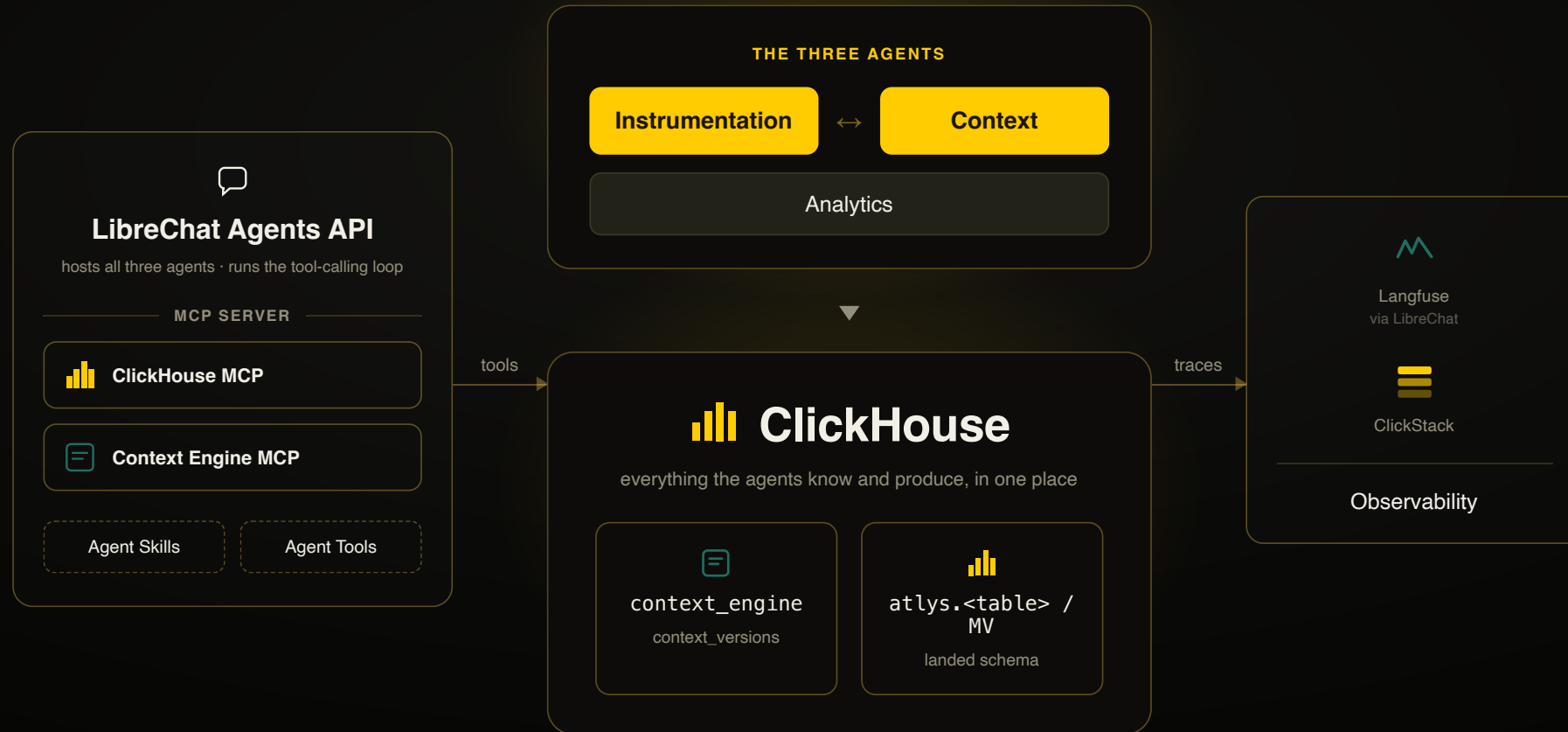
The Architecture

Feature spec in → schema, context, and insight out — three agents, one shared harness, built for the Atlys dataset.

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SYSTEM

One harness, three agents, one ClickHouse service



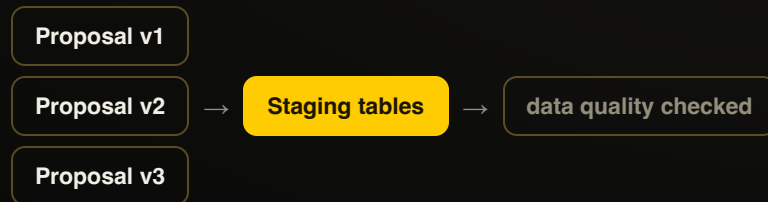
How do we ground schema generation?

An agent reasoning about a schema only from what it wrote down is reasoning about vibes, not the table. Two harnesses, used together.

Staging Harness

CREATED ONCE

A real ClickHouse staging area, provisioned once up front — every proposed schema is actually built and loaded with real data here before anyone trusts it.



USE CASE Every revision is tested for data quality here — not just re-argued in text.

Perf Harness

"This ORDER BY will be faster" is a guess, not evidence — so the winner is measured, never asserted.



USE CASE Times every candidate against staged data, flagging the agent if one is clearly the worst.

Where does the context engine live, and what does it hold?

If the agents' shared memory can quietly change, nothing built on top of it can be trusted.



Context Engine

ClickHouse · agent_meta

WHAT IT MODELS

Entities

Tables

Metrics

Relationships

Base context

SKILL



Context Engine CRUD

One consistent way for every agent to read and write context — no hand-rolled SQL per agent.

HOW WE STORE IT

context_versions

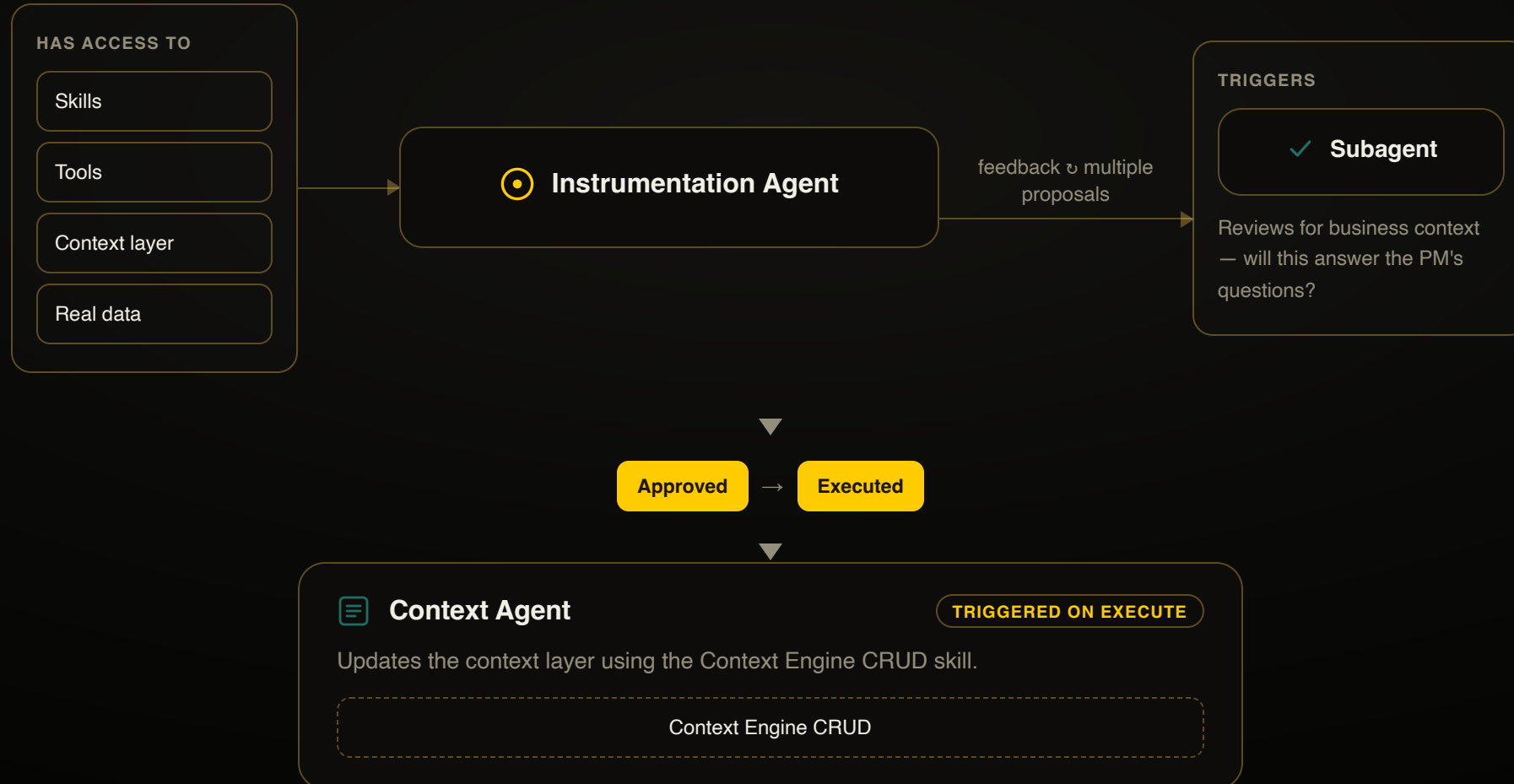
→ argMax(after, ts) →

current_context (view)

Append-only log, and a view over it — the latest write per section always wins, nothing is deleted.

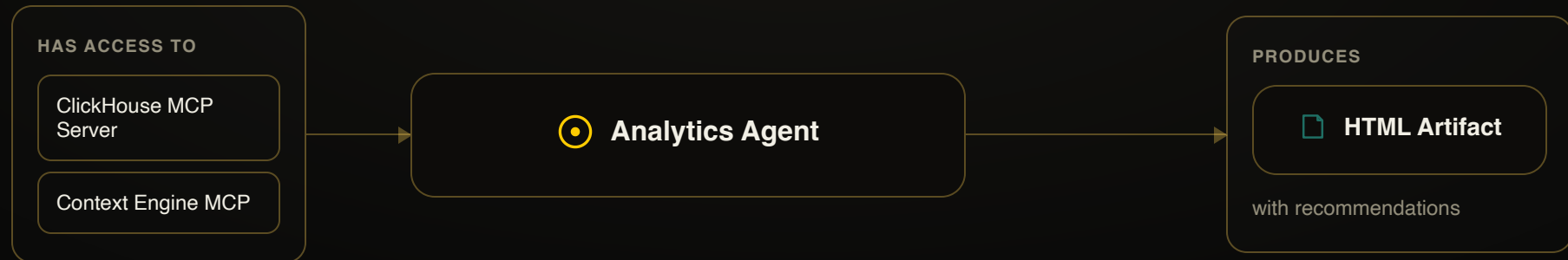
How does schema creation actually get decided?

One agent in the middle, with Skills, Tools, the Context Layer, and the real data — reasoning it out, not guessing it out.



What powers the Analytics Agent?

Same real access as the Instrumentation Agent — explores first, then writes the answer.



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Grounded, not guessed.

Visualize the architecture:

`claude.ai/code/artifact/8a54a040-667f-4921-9e4b-2b9e366443b5`

✱ LLM provider: **OpenAI** — cheapest tier, GPT-5.6 "Luna" — strong performance for the cost

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